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Advancing Safety Excellence: The Role of Coaching and Emerging Technologies in PDO HSE Development Programs

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Abstract

In the high-risk landscape of oil and gas operations, Health, Safety, and Environment (HSE) performance is no longer defined solely by compliance but by the adaptive capabilities of individuals and organizations. Petroleum Development Oman (PDO), Oman's leading oil and gas producer, has embarked on a transformative journey to elevate its safety culture by integrating structured coaching, digital technologies, and inclusive learning programs across its workforce and contractor base. This manuscript presents a comprehensive review of PDO's integrated HSE capability programs, particularly the Shababuna, Eidaad, Emdaad, and CSR coaching initiatives. Each initiative aligns with the 70-20-10 learning model and addresses different stages of professional development, ranging from students to entry-level professionals, mid-career workers, and contractors.

The manuscript explores how more than 40,000 Coachees have benefited from structured coaching and how more than 700 Omanis have been upskilled through developmental pipelines such as Shababuna, Eidaad, and Emdaad over the past years. The approach prioritizes human factors, psychological safety, and operational learning to create a resilient, forward-looking safety culture. PDO demonstrates a scalable and replicable strategy for achieving Goal Zero.

Introduction

As the energy sector undergoes rapid transformation, safety performance remains a critical benchmark of operational excellence. In upstream oil and gas environments characterized by technical complexity, high consequence potential, and distributed workforces, the ability to adapt and learn is central to preventing incidents and fostering resilience. Traditional training programs, often compliance focused and event based, have increasingly proven inadequate in sustaining long term behavioral change or fostering proactive safety mindsets (Reason, 1997; Hopkins, 2013).

Recognizing this shift, PDO has adopted a strategic and people centric approach to HSE capability development. This involves embedding coaching practices, leveraging digital innovation, and fostering partnerships to build HSE capacity across internal teams, contractors, and the wider national talent pool. With a strong focus on localization, inclusivity, and real-world engagement, PDO's programs aim to build a

safety conscious, adaptable, and high performing workforce aligned with Sultanate of Oman development goals and international safety standards.

At the heart of this transformation is the integration of the internationally recognized 70-20-10 learning model. In this model, 70% of learning is derived from on-the-job experience, 20% from coaching and mentoring, and 10% from formal training (Lombardo & Eichinger, 1996). PDO's implementation of this framework cuts across all its flagship programs, such as the Shababuna, Eidaad, Emdaad, and CSR coaching initiatives for Local Community Contractors (LCCs) and Small and Medium Enterprises (SMEs).

Literature Review

The Shift from Compliance to Capability

Modern safety management has seen a theoretical evolution from prescriptive, compliance driven paradigms to adaptive models focused on systems thinking and resilience. Scholars such as Hollnagel (2014), Dekker (2014), and Weick & Sutcliffe (2015) have emphasized the importance of human performance variability, learning in real time, and organizational mindfulness. In this context, HSE capability is no longer limited to rules and checklists but extends to what people Know, Say, and Do under conditions of uncertainty.

PDO operationalizes this shift through its "Know-Say-Do" capability model. This model underpins how safety behaviors are not just taught but internalized, communicated, and enacted within operational contexts. Such an approach aligns with the view of Reason (1997), who argues that safety is primarily a product of cultural and behavioral alignment, not procedural adherence alone.

Coaching as a Leadership Tool in HSE

Coaching, particularly in high-risk industries, is increasingly recognized as a critical enabler of proactive safety culture. Rather than relying on top-down enforcement, coaching facilitates reflective dialogue, ownership, and situational learning (Guldenmund, 2000; Zohar, 2010). PDO's HSE coaching framework aligns with best practices in professional coaching, including open-ended questioning, active listening, and avoiding directive advice. These methods encourage frontline workers and leaders to reflect, learn, and act with greater self-awareness.

International Coaching Federation (ICF) certified methodologies and frameworks from the Association of Coaching (AoC) were integrated into PDO's coaching upskilling plan.

Inclusive Talent Pipelines in HSE

Localization and workforce nationalization are essential elements of PDO's broader strategy. The Shababuna program was launched to accelerate the development of Omani graduates into technically and behaviorally competent professionals. Fairness in selection is ensured through standardized processes: IQ tests, GPA thresholds, and multilingual interview panels. To date, 782 graduates have completed the program across various disciplines, including a significant number from HSE department.

The Emdaad program, with a duration of 18 months, targets professionals through a structured certification path in collaboration with Oman Energy Association (OPAL) and Scottish Qualifications Authority (SQA). More than 400+ professionals have graduated from Emdaad, benefitting from a blend of classroom learning, field exposure, monthly coaching, and final evaluations.

Meanwhile, the Eidaad program places undergraduate students in a 10-month internship where they gain practical HSE experience before returning to their final academic year. Since its inception, it has onboarded 25 students. These programs ensure diversity, fairness, and future-readiness in Oman's HSE talent pipeline.

Community Coaching and CSR Integration

Beyond internal staff, PDO places strong emphasis on community inclusion through the CSR HSE coaching initiative, targeting Local Community Contractors (LCCs) and SMEs. 1300+ contractors from LCCs/SMEs were engaged through workshops, case studies, and on-the-job coaching. This program helps develop foundational HSE competencies in underrepresented segments, aligned with PDO's In-Country Value (ICV) strategy and Sultanate of Oman Vision 2040 goals.

The Role of Technology in HSE Capability

Technology has become a key enabler in PDO's Journey of Transformation. Learning Management Systems (LMS), digital portfolios, and real-time coaching analytics are employed to track performance, deliver modular learning, and reinforce accountability.

Through coaching database, monthly dashboards monitor attendance, progress, and competency acquisition. Coaching conversations are logged to detect emerging trends, knowledge gaps, and cultural dynamics that might impact HSE performance.

Recognition for these programs has come through several prestigious awards, including:

- Shababuna Program was recognized at the Abu Dhabi International Petroleum Exhibition and Conference (ADIPEC).
- Excellence in Implementing AI-Driven HR Solutions (Emdad Redeployment & Transferability System) received the Human Capital Development Excellence Award at the Global Higher Education Exhibition (GHEDEX) (2024)
- Excellence in Implementing Professional Coaching in the Workplace (Kafa'a Programme) received the Human Capital Development Excellence Award at the Global Higher Education Exhibition (GHEDEX) (2024)
- At the Future Skills & HRD Conference, PDO's 'Shababuna' program, focused on youth development, was recognized for its innovative use of technology and collaborative coaching and learning platforms (2023)
- CIPD (Chartered Institute of Personnel and Development) Award for Best Coaching & Mentoring Initiative (2024)

PDO's approach to HSE capability development reflects a paradigm shift in safety leadership, emphasizing human behavior, structured coaching, and technological enablement over traditional compliance measures. With more than 700 graduates across Shababuna, Eidaad, and Emdaad programs, and 40,000+ coaching beneficiaries to date, PDO is redefining how safety excellence can be achieved and sustained in complex industrial environments.

Program Design & Methodology

PDO has implemented a comprehensive and integrated suite of HSE coaching and development programs designed to build capability, reduce incidents, and instill a resilient safety culture across its operations and contractor ecosystem. This section outlines the structure, methodology, and rationale behind these programs, highlighting their alignment with the 70-20-10 learning model and PDO's strategic approach to human and operational excellence.

Embedding the 70-20-10 Learning Strategy in HSE Talent Development

PDO's HSE talent development approach is firmly rooted in the globally recognized 70-20-10 learning framework, which promotes a balanced and effective method of building capability. According to this model, 70% of learning occurs through hands-on, on-the-job experiences; 20% through social

learning, including peer collaboration, coaching, and mentorship; and the remaining 10% through formal education or structured training. This approach is widely adopted by leading global organizations and has proven particularly effective in driving sustainable behavioral change when combined with coaching and performance feedback (Jennings & Wargnier, 2010).

PDO's development programs Shababuna, Eidaad, Emdaad, and CSR Coaching are strategically designed to reflect and operationalize this model. Each program embeds learning directly into real operational contexts, enabling participants to acquire both technical and non-technical competencies through workplace immersion. For example, Shababuna provides graduates with a 3–5-year structured development journey focused primarily on experiential learning, reinforced by consistent coaching and formal modules. Similarly, Eidaad and Emdaad leverage structured field placements, feedback loops, and digital innovations to deliver scalable, measurable development outcomes. By anchoring its learning strategy in the 70-20-10 model, PDO ensures that talent development is not only comprehensive and practical but also aligned with its long-term HSE goals.

Programmatic Structure and Objectives

Shababuna: Structured Onboarding for Graduate Talent

The Shababuna Program serves as a flagship talent acceleration initiative for fresh university graduates. Designed to span 3 to 5 years, this structured onboarding journey develops both technical expertise and leadership acumen.

Participants undergo a rigorous selection process, which includes Intelligence Quotient (IQ) tests, Grade Point Average (GPA) screening, and multi-stage interviews, ensuring transparency and fairness. Upon successful enrollment, candidates are mapped onto one of 16 structured job tasks and progress through technical and behavioral competency frameworks.

The program culminates in the qualification of participants as Job Competency Profile (JCP). This structured, performance-based pathway is supplemented with coaching interventions and learning labs to reinforce PDO's safety philosophy and leadership principles.

Notably, the Shababuna Program was internationally recognized by the ADIPEC Awards in 2023, winning in the "Future Energy Workforce Development Programme" category. This recognition underscores its innovative use of collaborative learning platforms to enhance talent progression.

Eidaad: University Industry Bridge

Eidaad is a 10-months internship and development program designed to build HSE awareness and workplace readiness among university students. Students are selected through a merit-based system and placed in field operations, safety departments, and support functions within PDO and partner organizations.

The program unfolds in four key stages:

1. Talent Onboarding – where students undergo assessment and are provisionally accepted.
2. Industry Placement – involving official confirmations and potential interviews.
3. Industry Internship – comprising a 10-month immersive assignment.
4. Return to University – where students complete their studies with enhanced HSE awareness and practical insights.

This program equips young Omanis with foundational HSE knowledge and builds an early talent pipeline for future HSE professionals. As of mid-2025, 25+ students have completed this program, many of whom have pursued further roles in PDO or other energy sector firms.

Emdaad:

The Emdaad Program is an 18-month professional upskilling and nationalization initiative targeting Omanis. It includes six months of knowledge training and 12 months of on-the-job training (OJT) and has been active for over five years.

This program is underpinned by partnerships with OPAL (Oman Society for Petroleum Services), MOHERI (Ministry of Higher Education, Research and Innovation), ministry of Labour, and international certifying bodies such as SQA (Scottish Qualifications Authority). Over 500 professionals have benefited from Emdaad, with structured competency tracking, monthly coaching visits, and final assessments conducted by PDO and OPAL.

Candidates are supported through:

- Registration in national learning hubs (e.g., OPAL STAR)
- Portfolio building aligned with assessor and auditor criteria
- Monitoring via site visits and monthly performance reports
- Certification and post-program tracking of employment outcomes

This approach ensures long-term employability, practical capability, and alignment with national workforce development goals.

In 2024, the Emdaad Program received four awards at the GHEDEX Awards for Excellence in Human Capital Development, recognizing its talent systems and integration of workplace coaching (e.g., Engineering Skills Development Programme (KAFA'A)).

CSR Program: Developing HSE Capability in LCCs and SMEs

PDO's Corporate Social Responsibility HSE coaching program was created to build HSE capability in Local Community Contractors and Small and Medium Enterprises. These contractors often represent the lowest maturity segment in terms of HSE awareness and system development.

This initiative targets fresh graduates and young professionals with up to three years of experience, particularly those employed by subcontractors and local operators. Participants engage in interactive training formats including quizzes, case studies, site walkthroughs, and assessments.

Since inception:

- Over 1300+ contractors have participated in the CSR initiative.
- The coaching has improved compliance with PDO HSE protocols.
- Several participants have progressed into full-time HSE roles.

This initiative aligns directly with PDO's In-Country Value (ICV) Strategy, driving economic participation and improving operational consistency across PDO's supply chain. The program also won the 2025 Chairman's Award of Excellence for Upskilling HSE Capabilities for Local Community Contractors and SMEs and contributed to conference presentations by two program alumni in 2025.

Safety Support Center (SSC)

Alongside its coaching ecosystem, PDO formed the Safety Support Center (SSC) in 2014 to reinforce safety behaviors operationally. This centralized hub uses over 800 CCTV cameras to monitor real-time site activities across well operations. The SSC team analyzes live and recorded behaviors, deploying Behavioral Based Safety (BBS) interventions, conducting ergonomic reviews, and aiding incident investigations and

coaching in near real-time. In 2023 alone, the SSC recorded over 14,000 incidents on its proactive dashboard. The data has been used to:

1. Eliminate six recurrent unsafe behaviors.
2. Drive equipment design improvements and tailored safety belt-walk interventions.
3. Reduce Total Recordable Injury Rate (TRIR) by approximately 56% over five years.
4. Implement AI-driven hazard detection and predictive analytics for safer, proactive decision-making.

Further, SSC learnings are cascaded through PDO wide events and shared with contractor teams. Contractors are encouraged to adopt similar CCTV based behavior reviews. SSC success has inspired local oil and gas contractors across Sultanate of Oman to replicate its structure.

The SSC serves as a real-time behavioral reinforcement enabler, complementing coaching and formal instruction by delivering HSE insights on demand and reinforcing system-wide learning.

Integrated Coaching Support Framework

In parallel with these structured programs, PDO launched a dedicated coaching initiative in 2023, embedding 23 full-time certified HSE coaches across its operational clusters. The aim was to integrate coaching into daily workflows and elevate psychological safety, risk perception, and leadership communication across the enterprise.

As of June 2025:

- Over 3,000 coaching sessions have been delivered.
- More than 40,000 coachees have been engaged, including PDO staff and contractor personnel.
- The coaching framework includes 30 learning modules, focused on communication, influence, decision-making, risk recognition, and leadership behavior.

HSE Coaching Curriculum Overview

The HSE coaching curriculum currently includes 30 modules that are mandatory for all personnel in Safety Critical Positions. For other staff, completion of these modules is optional. Personnel holding positions responsible for designing, implementing, operating, or maintaining barriers, recovery preparedness measures, or escalation controls for managing hazards classified as Red Risk on PDO RAM or Yellow 5A/5B risks (Major Accident Hazards) are defined as Safety Critical positions.

Curriculum Delivery

Each module is delivered in a single session. Coaching acts as a means to facilitate intentional positive change in how HSE is implemented in the field. It primarily targets HSE practitioners and Business Leaders (OTE2 leaders) from PDO and contractors. The goal is to improve how these individuals execute and communicate HSE activities, while building their HSE knowledge.

Philosophy of HSE Coaching

HSE Coaching is about developing the capability of others. It is not focused on compliance, rules, or processes, but rather on how individuals perform the HSE aspects of their roles. For coaching purposes, HSE capability is divided into six categories:

- Base Knowledge and Context:
Knowledge of HSE theory and understanding of job-related HSE roles and responsibilities, focused mainly on HSE professionals but relevant to operations roles as well.

- **Control of Work:**
HSE professionals contribute to developing and implementing HSE management systems, ensuring work is planned, authorized, and managed safely. Business leaders own this process, ensuring HSE remains a priority.
- **Creating Influence:**
Since HSE professionals do not make operational decisions directly, they need to influence decisions to ensure safety outcomes.
- **Leading Change:**
Identifying and implementing HSE improvements requires managing change effectively, a responsibility shared by business leaders and HSE professionals.
- **Managing Risk:**
Both HSE professionals and business leaders must be skilled at assessing and managing risk as a core activity.
- **Facilitating Learning:**
Learning is not only from incidents but also from normal work activities to continuously improve safety performance.

Coaching Approach and Methods

HSE coaches are tasked with:

- Delivering group coaching sessions based on the 30 capability components to the HSE and operational community.
- Following up with one-on-one coaching sessions when requested by the coachee or when the coach identifies a need. These sessions use coaching prompts that include:
 - Things to look for (coaching opportunities)
 - Questions to ask to stimulate reflection and growth.

Coaches observe and engage with their cluster teams using these prompts as guidance. Coaching was designed to promote the shift from compliance-based management to capability-based leadership. Rather than enforcing checklists, coaches used structured conversation models rooted in behavioral science:

- Open-ended questions to stimulate self-reflection.
- Active listening and silence to allow coachees to process.
- Collaborative problem-solving without providing direct answers.
- Follow-up observations to reinforce learning in the field.

Some of the sessions were adapted for cluster specific risk contexts and workforce compositions, making the methodology scalable yet relevant.

By mid-2025, data indicated significant improvements in self-reported and observed behaviors:

- 30% of coachees reported greater confidence in speaking up about hazards.
- Supervisors observed increased toolbox talk participation and hazard reporting across major contractor sites.
- Near-miss reporting increased, particularly among previously under-reporting segments.

Supporting Digital Infrastructure and Monitoring

To manage scale and consistency, all coaching and development programs are supported by digital dashboards and analytics tools, including:

- Centralized tracking of coaching attendance, feedback scores, and progression status
- SSC Dashboard and AI Engine provides behavior alerts and analytic insights.

These tools enabled continuous program iteration, resource reallocation, and performance forecasting, improving responsiveness and targeting.

Additionally, participants from the Emdaad program were registered with SQA, OPAL, and internal PDO systems, allowing for seamless data integration, certification management, and career tracking.

Methodological Foundations

All programs and coaching interventions were grounded in key theoretical and empirical frameworks from the safety and adult learning literature:

- Reason's (1997) Human Error Model and Dekker's (2014) Safety Differently informed the focus on system resilience and human capability.
- Nielsen (2014) and Geller (2005) emphasized behavioral safety and participative learning models.
- The "Know, Say, Do" model served as the core capability framework, aligning knowledge (technical awareness), communication (safety conversations), and behavior (field application).

This structure allowed PDO to shift the learning paradigm from one-off compliance training to embedded capability-building, reinforcing safety ownership, reflective learning, and leadership growth across diverse segments of the workforce.

PDO's multi-program strategy including Shababuna, Eidaad, Emdaad, and CSR Coaching demonstrates a cohesive and evidence-based methodology for developing HSE capability and leadership at scale. By combining structured onboarding, coaching, digital enablement, and national development strategies, PDO has laid a strong foundation for cultural transformation and safety excellence across Sultanate of Oman's energy sector.

Results and Impact Assessment

Incident Reduction Metrics

When comparing 2021 to mid-2025, substantial declines were observed:

- Fatalities fell by over 70%, signaling effective intervention in life-critical processes.
- Lost Time Injuries (LTIs) declined by more than 70%.
- High Potential Incidents (HiPos) dropped by over 85%.
- Serious Injury or Fatality (SIF) events reached zero in two of the past three reporting years.

These outcomes reinforce that learning translates into safer behaviors and better hazard control.

Coaching Session Reach and Engagement

As referred in Figure 1, to date, the program has conducted approximately 3,000 coaching sessions, reaching over 40,000 participant attendances.

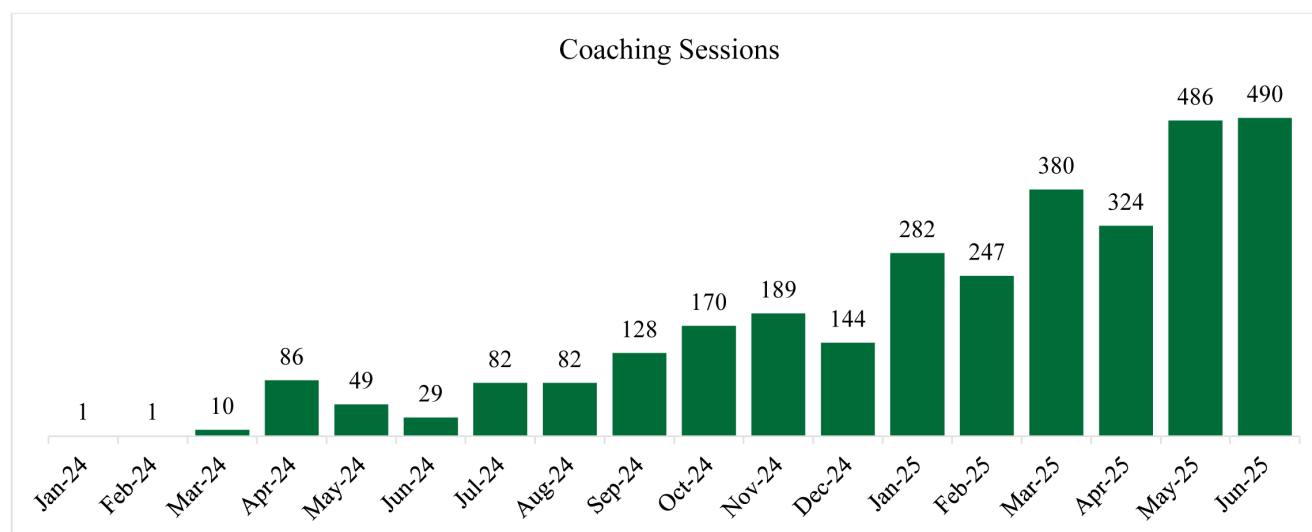


Figure 1—Distribution of Coaching Sessions / Month wise (PDO HSE Coaching Report, 2025)

Note: The reported 40,000 participants represent cumulative attendances across all coaching sessions, including repeat attendance by individuals attending multiple modules or refresher sessions. Most coaching sessions are conducted in groups averaging 15–20 participants, which enables a relatively small number of sessions to reach a large total number of participants. In addition to the base program, starting this year, we have launched the Coach the Coach initiative. In this program, experienced HSE advisors are coached to become coaches themselves, enabling them to support the coaching process, especially for contractors. This initiative is designed to expand coaching capacity by developing internal coaches across contractor organizations, ensuring coverage and sustainability of the HSE coaching program. Localized coaching clusters and flexible scheduling facilitated effective participation, integrating coaching into operational routines. Feedback from coaches emphasized that having accessible, on-site sessions contributed significantly to attendance and engagement.

Qualitative Insights and Feedback

Participant Surveys and Reflections

Open feedback themes included:

- "Now I look for safety proactively, not reactively."
- "The psychological safety module helped me speak up more confidently."
- "Coaching helped me understand not just what to do, but why."

Contractor Engagement and Testimonials

Interviews and thematic analysis showed:

- Major Contractor case study: 40% drop in recordable incidents and 150% rise in near-miss reports after 12 months of focused coaching.

- LCC/SME Worker testimonial: "The coaching sessions helped me see safety from a new perspective. It's not just about rules but how we engage and influence others."
- Supervisor feedback: "Having a HSE coach onsite made it easier to ask questions and improve daily HSE practices."

These examples underscore transformative changes beyond metric improvements.

Organization-Wide Coaching Support

PDO's formal HSE Coaching Program was officially launched in early 2024. However, a pilot phase began in 2021 with 2–3 coaches focusing on selected clusters and high-risk contractor groups. Since then, over the 4.5-year period (including pilot), the program has delivered more than 3,000 coaching sessions and engaged over 40,000 cumulative coaching participants. These figures include repeat engagements, as participants often attend multiple coaching modules or sessions.

HSE Coaching Stream (Full-Time Coaches)

- 23 full-time certified HSE coaches
- Recruited internationally and deployed onsite 12/7
- Dedicated exclusively to the HSE Coaching Program
- Focused on safety-critical roles and frontline operational teams
- Provide continuous, high-impact coaching aligned with PDO's Goal Zero objectives

Cultural Shifts and Systemic Enablement

Mindset and Psychological Safety

Coaching has enabled a shift in leadership conversations from "What went wrong?" toward "What could go wrong?", fostering anticipation of risk, shared accountability, and team-owned safety.

Discussion

PDO's approach to HSE capacity-building has bridged the gap between technical development and cultural transformation. By concurrently targeting internal leadership and external contractor populations including SMEs, LCCs, and university students, PDO demonstrated that safety improvement requires aligned capabilities across the organizational ecosystem, not isolated training or policy mandates.

The programs Shababuna, Eidaad, Emdaad, and the CSR Coaching Program serve distinct developmental stages. Each program reinforces PDO's overarching "Know, Say, Do" framework, building the knowledge, language, and behavior needed to embed safety into operational identity. For instance, while Shababuna develops young graduates in over 3–5 years through real-world task rotations, the Eidaad program places students into PDO environments before they graduate, promoting early professional socialization into safety practices. The Emdaad program accelerates professional certification and upskilling for over 400+ Omanis, supported by the OPAL STAR and SQA systems, while the CSR program advances safety leadership within underserved contractor communities.

This systematic approach reflects the broader logic of the 70-20-10 learning model that PDO has embraced: 70% experiential (on-the-job), 20% social (coaching and mentorship), and 10% formal (courses and certifications). Instead of viewing HSE development as event-based training, PDO has redefined it as a continuous, embedded process, facilitating not only procedural competence but psychological ownership.

Coaching as a Cultural Lever

One of the most transformative aspects of PDO's strategy is the embedded coaching model. Coaching has moved beyond instruction into a method of thinking and leading. Coaches emphasize open-ended questions, reflection, and dialogue rather than prescriptive directives. By avoiding top-down enforcement and instead fostering collaborative discovery, they have enhanced both trust and capability.

Coaching Sessions and Participant Breakdown

[Table 1](#), demonstrates the extensive participation of PDO's contractor and internal leadership cohorts in the coaching program. Major contractors represent the largest portion of both sessions and coachees, reflecting their significant operational presence. The inclusion of LCCs and SMEs, although smaller in absolute numbers, shows targeted efforts to engage a diverse contractor base. PDO's Safety Critical Positions account for nearly half of the sessions and a quarter of the coachees, underscoring the focus on leadership development.

Table 1—Distribution of Coaching Sessions and Participants (PDO HSE Coaching Report, 2025)

Population Group	Number of Group and One-to-One Coaching Sessions	% of Total Coaching Sessions	Number of Coachees	% of Total Coachees
PDO Major Contractors	1,500+	50%	30,000+	75%
LCCs & SMEs	30+	1%	1,300+	3.25%
PDO Safety Critical Positions	1,400+	46.7%	10,000+	25%
Total	3,000+	100%	40,000+	100%

Coaching has allowed PDO to shift leadership behavior from compliance enforcement to guidance, reinforcing [Reason's \(1997\)](#) shift from a person-centric to a system-centric understanding of safety. Leaders were not just trained they were coached to ask better questions, to listen, and to model safe behavior. This shift has resulted in stronger psychological safety, more team-led safety initiatives, and a measurable increase in speaking-up behavior, especially among younger and lower-level workers. This is consistent with [Weick and Sutcliffe's \(2015\)](#) theory on High Reliability Organizations (HROs), where mindfulness and deference to expertise at the frontline are essential.

Moreover, PDO's coaching methodology is informed by adult learning theory ([Knowles, 1984](#)), which emphasizes autonomy, relevance, and practical application. Coaches function as facilitators who support adult learners in constructing their own understanding, making the learning "stick" far more effectively than traditional compliance-based training.

Contractor Inclusion and In-Country Value

The inclusion of major Contractors, local SMEs and LCCs reflects a strategic integration of HSE improvement with PDO's ICV goals. Historically, contractor populations face systemic challenges resource constraints, limited HSE knowledge, and lower maturity in HSE systems. Yet through dedicated coaching, quiz-based learning, and contextual case studies, PDO raised the safety awareness and ownership of over 1300 LCCs and SMEs.

This initiative served multiple purposes. First, it elevated HSE practices in a high-risk, under-regulated segment of PDO's supply chain. Second, it promoted inclusive development by ensuring that safety excellence was not a privilege of the core operator workforce but a shared responsibility. Third, it created a pipeline of competent, safety-conscious professionals who could be employed across Oman's broader

industrial landscape, not just within PDO. By aligning social impact with business outcomes, PDO strengthened both safety and sustainability.

Leadership Development and Recognition

PDO's programs have not only improved internal HSE performance but have been widely acknowledged. The Shababuna program, which fast-tracked the development of over 782 young Omanis and used analytics for training needs, won the prestigious ADIPEC Award in 2023. The Emdaad program's change management and coaching innovations won four separate awards at the GHEDEX Excellence Awards. This recognition affirms PDO's commitment to innovation, continuous improvement, and operational excellence.

Limitations

Attribution and Longitudinal Causality

PDO continues to drive safety and performance excellence through a diverse portfolio of complementary initiatives, including Boots on Ground (BoG), Musta'ed, Ihtimam, HSE Mindset, Human Performance Improvement (HPI), Lean/CI methodologies etc., These programs work in tandem, reinforcing shared values, behaviors, and technical improvements across the organization. While the observed reductions in incidents and improvements in safety culture are highly encouraging, it is important to acknowledge that such outcomes result from the collective impact of multiple integrated efforts. Attributing success to any single initiative would not fully reflect the systemic nature of change underway. Recognizing this, PDO is now exploring longitudinal studies and data-driven approaches to better understand the distinct and combined contributions of coaching and other interventions further strengthening our commitment to continuous learning and evidence-based improvement.

Data Integrity and Reporting Bias

Although the coaching database and coaching feedback tools are robust, the quality of data depends on honest feedback, consistent logging, and coach engagement. There is always potential for reporting bias, especially in self-assessment tools or when coaching quality varies between individuals.

Variable Coaching Quality

Not all coaches are equally experienced or effective. While the ICF and AoC frameworks provide strong foundations, delivery styles and contextual knowledge can vary. This variation may affect participant engagement and outcomes. PDO is addressing this through peer reviews, supervision models, and ongoing professional development.

Contractor Heterogeneity

The diversity of LCCs ranging from micro to super SMEs means that one-size-fits-all coaching or training modules are not always appropriate. Tailoring interventions for resource-limited contractors requires high effort and flexibility, challenging program scalability without compromising quality.

Time Constraints and Operational Demands

Operational sites are under constant pressure to deliver results. Scheduling coaching sessions amidst demanding production schedules sometimes creates friction or limits full participation, especially among contractor teams. PDO has mitigated this by embedding coaching into routine workflows and encouraging "coaching on the go," but the issue remains.

Recommendations

Longitudinal Studies and Benchmarking

To validate long-term impact, PDO should invest in multi-year longitudinal studies comparing coached and uncoached populations. Benchmarking against regional and international operators would provide useful insights. Control groups, while challenging, could be considered in academic partnerships.

Advance Coach Certification and Supervision

Continuous development for internal coaches is vital. PDO is considering master-coach tracks, periodic re-certification, and reflective supervision models. Partnering with global coaching institutions for train-the-trainer programs could ensure consistency and quality.

Formalize Coaching as a Leadership Competency

PDO has plans to embed coaching into leadership competency frameworks through Sadara Program. Leaders who demonstrate coaching ability especially in safety-critical areas should be rewarded and spotlighted. This would further entrench coaching as a strategic pillar of leadership.

Conclusion

The discussion reinforces that PDO's approach is more than a training initiative it is a transformation agenda. By combining coaching, structured development programs, digital infrastructure, and inclusive outreach, PDO is building a future-ready, resilient safety culture. Despite limitations in attribution and data variance, the scale of change 40,000 coached, 700+ graduates, incident reductions exceeding 70% points to a robust, evidence-based shift.

PDO's success offers a replicable model for the oil and gas sector: safety excellence anchored in human capability, social learning, and technological enablement. The next phase must focus on deepening impact, expanding inclusion, and sustaining behavior change long after the coaching session ends.

What sets PDO apart is its deliberate shift from a compliance-driven model to a capability-centered one. The traditional model of safety management often stops at policy adherence and procedural audits. PDO's model goes further: embedding safety as a mindset, not just a rulebook. This is evident in the increased quality of conversations in the field, improved near-miss reporting, and the shift from "what went wrong" to "what could go wrong," demonstrating proactive risk anticipation.

Together, these outcomes affirm PDO's position as a regional and international leader in workforce development and HSE innovation.

Future Directions

Despite its achievements, PDO recognizes that transformation is an ongoing process. Maintaining momentum and deepening impact requires adaptive learning, scaling inclusive practices, and anticipating emerging industry risks. The following future-focused strategies outline how PDO intends to continue advancing HSE excellence.

Expanded Contractor and Informal Workforce Inclusion

While over 1300+ LCCs have benefited from CSR coaching, PDO plans to reach a broader segment of Oman's contracting and informal workforce. Special emphasis will be placed on micro-enterprises and informal subcontractors, who often lack structured safety systems. Customized modules will be created

using low-bandwidth, mobile-accessible formats in Arabic, and coaching delivery will be adapted to local contexts with community-based facilitators.

PDO will also increase integration with other national capacity-building platforms to ensure alignment and reduce duplication of effort across ministries and industry bodies.

Longitudinal and Comparative Evaluation

One of the main limitations acknowledged in the discussion was the difficulty in attributing improvements solely to coaching. To address this, PDO will initiate longitudinal research studies in collaboration with academic institutions, following cohorts of coached and uncoached individuals over 3–5 years. These studies will measure not only safety performance metrics (e.g., SIF reductions, incident rates) but also behavioral and cultural indicators (e.g., psychological safety, team dynamics, safety ownership).

Control groups will be explored where feasible, and benchmarking against other Middle East and international operators will provide context and validation.

Formalizing Coaching in Career Progression

To institutionalize coaching as a leadership competency, PDO will embed coaching skills into its performance management and promotion frameworks. Leaders who demonstrate effectiveness in developing others, fostering safety conversations, and guiding teams through complex decisions will be formally recognized and rewarded.

Training curricula for first-time supervisors, frontline managers, and senior leaders will all include coaching as a core capability, reinforcing its role as a leadership norm rather than a specialized activity.

Knowledge Sharing and Global Collaboration

Recognizing its role as a regional leader, PDO will open its coaching models, modules, and impact data to cross-industry collaboration. Through workshops, conferences, and partnerships with SPE, OPAL, and IOGP, PDO aims to contribute to the global body of knowledge on HSE coaching and behavioral safety.

Strategic Reflections

PDO's success demonstrates that HSE improvement is not merely a technical challenge it is a human systems challenge. Traditional training, while important, does not suffice to change mindsets or build a culture of vigilance. By prioritizing capability over compliance, coaching over command, and inclusion over isolation, PDO has advanced a new frontier in operational safety and workforce empowerment.

The vision is clear: to create a future-ready, adaptive, and psychologically safe workforce, capable of anticipating risks, responding to uncertainty, and continually learning. This cannot be achieved by rules alone. It requires leaders who coach, teams who trust, and systems that learn.

PDO's multi-layered approach rooted in evidence, supported by technology, and animated by values offers a replicable model for others in the oil and gas sector and beyond.

Final Remarks

As PDO continues this journey, it does so with humility, recognizing the evolving nature of HSE risks and the need for continuous learning. At the same time, it offers inspiration: that even in the high-risk, high-stakes environments of oil and gas, it is possible to foster a culture of care, competence, and courage where every individual, from graduate trainee to subcontractor, becomes a steward of safety.

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